

OOPS Assignment 05

Code:

```
import javax.swing.*;
import javax.swing.table.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class LanguageInstitute extends JFrame {
    JTextField
    txtId,txtName,txtLevel,txtBatch,txtFees,txtTrainer,txtContact,txtPerformance;
    JComboBox<String> cmbLang;
    JCheckBox chkExam;
    JTable table; DefaultTableModel model;
    Connection con;

    LanguageInstitute() {
        if (!showLogin()) System.exit(0);
        connectDB();
        setTitle("Language Institute"); setSize(1200,700);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLayout(new BorderLayout());

        // Form Panel
        JPanel form = new JPanel(new GridLayout(0,2,5,5));
        form.setBorder(BorderFactory.createEmptyBorder(10,10,10,10));
        form.setPreferredSize(new Dimension(280,0));
        txtId=tf(); txtName=tf(); txtLevel=tf(); txtBatch=tf();
        txtFees=tf(); txtTrainer=tf(); txtContact=tf(); txtPerformance=tf();
        cmbLang=new JComboBox<>(new
String[]{"English","French","Russian","Hindi","German","Spanish","Japanese"});
        chkExam=new JCheckBox("Exam Enrolled");

        String[]
        labels={"ID","Name","Language","Level","Batch","Fees","Trainer","Contact","Performanc
e","Exam"};
```

```

JComponent[]
fields={txtId,txtName,cmbLang,txtLevel,txtBatch,txtFees,txtTrainer,txtContact,txtPerform
ance,chkExam};
for(int i=0;i<labels.length;i++){form.add(new JLabel(labels[i]));form.add(fields[i]);}

// Buttons
JPanel btns=new JPanel(new FlowLayout());
for(String[] b:new
String[][]{{"Add","add"},{"Update","upd"},{"Delete","del"},{"Search","srch"},{"Reset","rst"},{
"Exit","exit"}}){
    JButton btn=new JButton(b[0]);
    btn.setActionCommand(b[1]);
    btn.addActionListener(this::handle);
    btns.add(btn);
}
JPanel left=new JPanel(new BorderLayout());
left.add(form,BorderLayout.CENTER); left.add(btns,BorderLayout.SOUTH);

// Table
model=new DefaultTableModel(new
String[]{"ID","Name","Lang","Level","Batch","Fees","Trainer","Contact","Perf","Exam","R
emarks"},0){
    public boolean isCellEditable(int r,int c){return false;}
};
table=new JTable(model); table.setRowHeight(24);
table.addMouseListener(new MouseAdapter(){
    public void mouseClicked(MouseEvent e){
        int r=table.getSelectedRow(); if(r<0)return;
        txtId.setText(model.getValueAt(r,0).toString());
        txtName.setText(model.getValueAt(r,1).toString());
        cmbLang.setSelectedItem(model.getValueAt(r,2));
        txtLevel.setText(model.getValueAt(r,3).toString());
        txtBatch.setText(model.getValueAt(r,4).toString());
        txtFees.setText(model.getValueAt(r,5).toString());
        txtTrainer.setText(model.getValueAt(r,6).toString());
        txtContact.setText(model.getValueAt(r,7).toString());
        txtPerformance.setText(model.getValueAt(r,8).toString());
        chkExam.setSelected("Yes".equals(model.getValueAt(r,9)));
    }
});

```

```

        add(left, BorderLayout.WEST);
        add(new JScrollPane(table), BorderLayout.CENTER);
        loadTable(); setLocationRelativeTo(null); setVisible(true);
    }

    JTextField tf(){return new JTextField();}

    void handle(ActionEvent e){
        switch(e.getActionCommand()){
            case "add" -> addRecord(); case "upd" -> updateRecord();
            case "del" -> deleteRecord(); case "srch" -> searchRecord();
            case "rst" -> resetFields();

        case "exit" -> {if(JOptionPane.showConfirmDialog(this, "Exit?", "Confirm", JOptionPane.YES_NO_OPTION) == 0) System.exit(0);}
        }
    }

    void connectDB(){
        try{Class.forName("com.mysql.cj.jdbc.Driver");

        con=DriverManager.getConnection("jdbc:mysql://localhost:3306/language_institute", "root", "");
        }catch(Exception e){JOptionPane.showMessageDialog(this, "DB Error: "+e.getMessage(), "Error", JOptionPane.ERROR_MESSAGE);}
    }

    void addRecord(){
        if(txtId.getText().isBlank() || txtName.getText().isBlank()){msg("ID and Name required."); return;}
        try{PreparedStatement ps=con.prepareStatement("INSERT INTO enrollment VALUES(?,?,?,?,?,?,?,?,?,?,?)");
            setParams(ps); ps.setString(12, ""); ps.executeUpdate(); msg("Added!");
            loadTable(); resetFields();
        }catch(Exception e){msg("Error: "+e.getMessage());}
    }

    void updateRecord(){
        if(txtId.getText().isBlank()){msg("Select a record."); return;}

```

```

        try{PreparedStatement ps=con.prepareStatement(
            "UPDATE enrollment SET
student_name=?,language_selected=?,course_level=?,batch_timing=?,fees_paid=?,trai
ner_name=?,contact_number=?,performance_status=?,certification_exam=?,remarks=?
,profile_path=? WHERE student_id=?");
            ps.setString(1,txtName.getText());
ps.setString(2,cmbLang.getSelectedItem().toString());
            ps.setString(3,txtLevel.getText()); ps.setString(4,txtBatch.getText());

ps.setDouble(5,txtFees.getText().isBlank()?0:Double.parseDouble(txtFees.getText()));
            ps.setString(6,txtTrainer.getText()); ps.setString(7,txtContact.getText());
            ps.setString(8,txtPerformance.getText());
ps.setString(9,chkExam.isSelected()?"Yes":"No");
            ps.setString(10,""); ps.setString(11,"");
ps.setInt(12,Integer.parseInt(txtId.getText()));
            ps.executeUpdate(); msg("Updated!"); loadTable();
        }catch(Exception e){msg("Error: "+e.getMessage());}
    }

    void deleteRecord(){
        if(txtId.getText().isBlank()){msg("Select a record.");return;}
        if(JOptionPane.showConfirmDialog(this,"Delete ID
"+txtId.getText()+"?", "Confirm",JOptionPane.YES_NO_OPTION)!=0)return;
        try{PreparedStatement ps=con.prepareStatement("DELETE FROM enrollment
WHERE student_id=?");
            ps.setInt(1,Integer.parseInt(txtId.getText())); ps.executeUpdate();
msg("Deleted!"); loadTable(); resetFields();
        }catch(Exception e){msg("Error: "+e.getMessage());}
    }

    void searchRecord(){
        if(txtId.getText().isBlank()){msg("Enter ID.");return;}
        try{PreparedStatement ps=con.prepareStatement("SELECT * FROM enrollment
WHERE student_id=?");
            ps.setInt(1,Integer.parseInt(txtId.getText())); ResultSet rs=ps.executeQuery();

if(rs.next()){txtName.setText(rs.getString(2));cmbLang.setSelectedItem(rs.getString(3));

txtLevel.setText(rs.getString(4));txtBatch.setText(rs.getString(5));txtFees.setText(rs.getStr
ing(6));

```

```

        txtTrainer.setText(rs.getString(7));txtContact.setText(rs.getString(8));

txtPerformance.setText(rs.getString(9));chkExam.setSelected("Yes".equals(rs.getString(
10)));
    }else msg("Not found.");
    }catch(Exception e){msg("Error: "+e.getMessage());}
}

void setParams(PreparedStatement ps) throws Exception {
    ps.setInt(1,Integer.parseInt(txtId.getText())); ps.setString(2,txtName.getText());
    ps.setString(3,cmbLang.getSelectedItem().toString());
ps.setString(4,txtLevel.getText());
    ps.setString(5,txtBatch.getText());

ps.setDouble(6,txtFees.getText().isBlank()?0:Double.parseDouble(txtFees.getText()));
    ps.setString(7,txtTrainer.getText()); ps.setString(8,txtContact.getText());
    ps.setString(9,txtPerformance.getText());
ps.setString(10,chkExam.isSelected()? "Yes": "No");
    ps.setString(11,"");
}

void loadTable(){
    try{model.setRowCount(0);
        ResultSet rs=con.createStatement().executeQuery("SELECT * FROM
enrollment");
        while(rs.next())model.addRow(new
Object[]{rs.getInt(1),rs.getString(2),rs.getString(3),
            rs.getString(4),rs.getString(5),rs.getDouble(6),rs.getString(7),rs.getString(8),
            rs.getString(9),rs.getString(10),rs.getString(11)});
    }catch(Exception e){msg("Load error: "+e.getMessage());}
}

void resetFields(){
    for(JTextField f:new
JTextField[]{txtId,txtName,txtLevel,txtBatch,txtFees,txtTrainer,txtContact,txtPerformance
})f.setText("");
    cmbLang.setSelectedIndex(0); chkExam.setSelected(false); table.clearSelection();
}

boolean showLogin(){

```

```

        JTextField user=new JTextField(); JPasswordField pass=new JPasswordField();
        JPanel p=new JPanel(new GridLayout(2,2,10,10));
        p.add(new JLabel("Username"));p.add(user);p.add(new
JLabel("Password"));p.add(pass);

if(JOptionPane.showConfirmDialog(null,p,"Login",JOptionPane.OK_CANCEL_OPTION,
JOptionPane.PLAIN_MESSAGE)!=0)return false;
    if(user.getText().equals("admin")&&new
String(pass.getPassword()).equals("1234")){msg("Login Successful!");return true;}
        JOptionPane.showMessageDialog(null,"Invalid credentials.","Login
Failed",JOptionPane.ERROR_MESSAGE);
        return false;
    }

    void msg(String s){JOptionPane.showMessageDialog(this,s);}
    public static void main(String[]
args){SwingUtilities.invokeLater(LanguageInstitute::new);}
}

```

SQL DATABASE Query:

```

CREATE DATABASE language_institute;
USE language_institute;

```

```

CREATE TABLE enrollment (
    student_id    INT PRIMARY KEY,
    student_name  VARCHAR(100),
    language_selected VARCHAR(50),
    course_level  VARCHAR(50),
    batch_timing  VARCHAR(50),
    fees_paid     DOUBLE,
    trainer_name  VARCHAR(100),
    contact_number VARCHAR(15),
    performance_status VARCHAR(50),
    certification_exam VARCHAR(5),
    remarks       TEXT,
    profile_path   VARCHAR(255)
)

```

Output:

Language Institute
Enrollment Management System

STUDENT INFO

Student ID

Full Name

Language

Course Level

Batch Timing

Fees Paid (₹)

TRAINER & STATUS

Trainer Name

Contact Number

Performance

Enrollment Records — 3 students

Click any row to load into form

ID	Name	Language	Level	Batch	Fees (₹)	Trainer	Contact	Performance	Exam	Remarks
1	Soham Pendse	German	2	11-1	1000.0	Hitler	1213131	Perfect for the h...	Yes	
2	Walsh D'Souza	Russian	4	2-3	4000.0	Joseph Stalin	31314	Communist	No	
3	Shirsh Shirodkar	Spanish	3	12-1	3000.0	Shrica Ma'am	1234567890	Great	No	

Add Student

Update

Delete

Search

Reset Form

Exit

phpMyAdmin

Recent Favorites

New

information_schema

language_institute

New

enrollment

mysql

performance_schema

phpmyadmin

test

Server: 127.0.0.1 » Database: language_institute » Table: enrollment

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

Triggers

Showing rows 0 - 2 (3 total, Query took 0.0017 seconds.)

SELECT * FROM `enrollment`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

	student_id	student_name	language_selected	course_level	batch_timing	fees_paid	trainer_name	contact_number	performance_status	certification			
☐	Edit	Copy	Delete	1	Soham Pendse	German	2	11-1	1000	Hitler	1213131	Perfect for the humanity	Yes
☐	Edit	Copy	Delete	2	Walsh D'Souza	Russian	4	2-3	4000	Joseph Stalin	31314	Communist	No
☐	Edit	Copy	Delete	3	Shirsh Shirodkar	Spanish	3	12-1	3000	Shrica Ma'am	1234567890	Great	No

Check all

With selected:

Edit

Copy

Delete

Export

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

The screenshot displays a web application titled "Language Institute Enrollment Management System". The interface is divided into a sidebar on the left and a main content area on the right.

Sidebar (Left):

- Language Institute**
Enrollment Management System
- STUDENT INFO**
 - Student ID: 2
 - Full Name: Walsh D'Souza
 - Language: (Dropdown menu)
 - Course Level: 4
 - Batch Timing: 2-3
 - Fees Paid (₹): 4000.0
- TRAINER & STATUS**
 - Trainer Name: Joseph Stalin
 - Contact Number: 31314
 - Performance: Communit

Main Content Area (Right):

- Enrollment Records** — 3 students
- Click any row to load into form

ID	Name	Language	Level	Batch	Fees (₹)	Trainer	Contact	Performance	Exam	Remarks
1	Soham Pendse	German	2	11-1	1000.0	Hitler	1213131	Perfect for the h...	Yes	
2	Walsh D'Souza	Russian	4	2-3	4000.0	Joseph Stalin	31314	Communist	No	
3	Shirsh Shirodkar	Spanish	3	12-1	3000.0	Shrica Ma'am	1234567890	Great	No	

Confirmation Dialog (Center):

Confirm Delete

Delete Student ID 2?

Yes No

Footer (Bottom):

Language Institute Enrollment System • Connected to localhost:3306

Buttons: Add Student, Update, Delete, Search, Reset Form, Exit


Language Institute — Enrollment System

File Edit Help



Language Institute
Enrollment Management System

LIVE

MySQL Connected

STUDENT INFO

Student ID

1

Full Name

Soham Pendse

Language

Course Level

2

Batch Timing

11-1

Fees Paid (₹)

1000.0

TRAINER & STATUS

Trainer Name

Hitler

Contact Number

1213131

Performance

Perfect for the humanity

Enrollment Records — 2 students

Click any row to load into form

ID	Name	Language	Level	Batch	Fees (₹)	Trainer	Contact	Performance	Exam	Remarks
1	Soham Pendse	German	2	11-1	1000.0	Hitler	1213131	Perfect for the h...	Yes	
3	Shirsh Shirodkar	Spanish	3	12-1	3000.0	Shrica Ma'am	1234567890	Great	No	

Updated

Record updated.

OK

Add Student

Update

Delete

Search

Reset Form

Exit

Language Institute Enrollment System • Connected to localhost:3306

phpMyAdmin

Recent Favorites

- New
- information_schema
- language_institute
 - New
 - enrollment
- mysql
- performance_schema
- phpmyadmin
- test

Server: 127.0.0.1 » Database: language_institute » Table: enrollment

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Showing rows 0 - 1 (2 total, Query took 0.0012 seconds)

SELECT * FROM `enrollment`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		student_id	student_name	language_selected	course_level	batch_timing	fees_paid	trainer_name	contact_number	performance_status	certification
☐	Edit Copy Delete	1	Soham Pendse	Japanese	2	11-1	1000	Hitler	1213131	Perfect for the humanity	Yes
☐	Edit Copy Delete	3	Shirsh Shirodkar	Spanish	3	12-1	3000	Shrica Ma'am	1234567890	Great	No

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None